

Topic -1: Introduction to Reinforcement Learning

AIMLCZG512-Deep Reinforcement Learning

Lecture Notes

1. Topics Covered

1. Formal meaning of reinforcement learning and the agent-environment view.
2. Difference between reinforcement learning, supervised learning, and unsupervised learning.
3. Core elements of reinforcement learning: policy, reward signal, value function, and model.
4. Typical application areas where sequential decisions and delayed consequences matter.
5. Review of key terms required for reading the first chapter of Sutton and Barto.
6. Tic-tac-toe as a toy example for value-based learning and temporal-difference updating.
7. Review questions for examination preparation.

2. What is Reinforcement Learning?

Reinforcement learning is a computational approach to **goal-directed learning from interaction**. The learner is called the **agent**; the external system with which it interacts is the **environment**. The agent observes the present situation, selects an action, receives a numerical reward, and then observes the next situation.

Formal textbook idea. Reinforcement learning is learning what to do, that is, how to map situations to actions, so as to maximize a numerical reward signal. The learner is not directly told which action is correct; it must discover useful actions by trying them.

At time step t , the agent observes a state $S_t \in \mathcal{S}$ and selects an action $A_t \in \mathcal{A}$. The environment then produces a reward $R_{t+1} \in \mathcal{R}$ and a new state S_{t+1} . The elementary interaction pattern is

$$S_t \longrightarrow A_t \longrightarrow (R_{t+1}, S_{t+1}).$$

The objective is to learn a way of acting that maximizes **cumulative reward over time**, not only the next immediate reward. Hence, reinforcement learning is not merely prediction. It is about choosing actions that influence future situations and future rewards.

2.1. Agent-Environment View

The agent-environment boundary is chosen based on what the learner can directly control. The agent controls its actions. The environment includes everything that responds to those actions, including the state transitions and the reward mechanism. A learning agent must be able to sense the state of the environment, take actions that affect that state, and have a goal expressed through reward.

The two most important distinguishing features are **trial-and-error search** and **delayed reward**. Trial and error is required because the agent must test actions to learn their consequences. Delayed reward is important because an action may affect the next state and, through it, many later rewards. This makes RL a natural framework for sequential decision-making under uncertainty.

3. How RL Differs from Supervised and Unsupervised Learning

Learning type	Learning signal	Main nature of the problem
Supervised learning	Labeled examples are provided by an external teacher. Each example includes an input and the desired output.	The learner generalizes from known examples to unseen examples. The feedback is instructive because it tells what the correct output should be.
Unsupervised learning	No reward signal and no labeled target are provided.	The learner attempts to discover structure in data, such as clusters, associations, patterns, or useful representations.
Reinforcement learning	A scalar reward is received as a consequence of actions.	The learner must interact with the environment, try actions, and improve future action selection for long-run reward. The feedback is evaluative rather than instructive.

In supervised learning, examples of correct behaviour are supplied. In reinforcement learning, such examples are usually not available. The agent may know that an outcome was good or bad, but it may not know which action should have been chosen instead. This creates the need for **exploration** and **credit assignment**. In unsupervised learning, the usual aim is to find hidden structure in data; in reinforcement learning, the aim is to learn behaviour that maximizes reward.

3.1. Important Characteristics of RL

- **Reward-based feedback:** RL uses rewards rather than direct labels for correct actions. The feedback evaluates the action taken, but it does not directly specify the best possible action.
- **Sequential decision-making:** The present action can change the next state, and the next state changes what actions are available later. Hence, a good decision must consider future consequences.
- **Delayed consequences:** Rewards may arrive much later than the action that caused them. A move that looks weak immediately may set up a strong future position. Similarly, a tempting immediate reward may lead to poor long-term return.
- **Exploration and exploitation:** The agent must use what it already knows, but it must also try less-known actions to discover whether better choices exist.
- **Uncertainty:** The agent is normally assumed to operate with incomplete knowledge of the environment. Outcomes may be stochastic, and the same action may not always lead to the same result.

Exploration-exploitation issue. To obtain reward, the agent should choose actions that already look good. To discover better actions, it must sometimes try actions whose values are not yet well known. If the agent only exploits, it may settle too early for a suboptimal action. If it only explores, it may fail to benefit from what it has already learned. Good reinforcement learning therefore requires a practical balance between both.

4. Elements of Reinforcement Learning

Beyond the agent and the environment, a standard reinforcement learning system has four main subelements: **policy**, **reward signal**, **value function**, and, optionally, a **model** of the environment.

Element	Meaning and role
Agent	The learner and decision-maker. It selects actions and improves its behaviour through experience.
Environment	Everything outside the agent that responds to actions and returns next states and rewards.
Policy	The agent's behaviour rule. It maps perceived states to actions. A deterministic policy gives one action; a stochastic policy gives action probabilities.
Reward signal	The immediate scalar feedback that defines what is good or bad for the agent. It is the primary basis for changing behaviour.
Value function	An estimate of long-term desirability. It tells how good a state, or a state-action pair, is by considering expected future reward.
Model	A predictive description of the environment. Given a state and action, a model may predict the next state and reward. It supports planning.

4.1. Policy

A **policy** defines the agent's way of behaving at a given time. It maps states to actions, or to probabilities over actions. The policy is central because it is sufficient to determine behaviour: once the current state and policy are known, the agent's action-selection rule is specified.

4.2. Reward Signal

The **reward signal** defines the immediate goal of the RL problem. It tells the agent what is desirable now. If an action is followed by low reward, the policy may later be changed so that a different action is selected in similar situations.

4.3. Value Function

Reward tells what is immediately good. **Value** tells what is good in the long run. A state may give low immediate reward but still have high value if it commonly leads to better future states. Conversely, a state may give immediate benefit but lead to poor future outcomes. Thus, RL decisions are usually based on value judgments, not only on immediate rewards.

4.4. Model

A **model** mimics the behaviour of the environment. Given a state and action, it may predict the next state and reward. Models support planning, where the agent considers possible future situations before actually experiencing them. Methods using such models are called **model-based methods**; methods learning directly from experience without an explicit model are called **model-free methods**.

5. Applications of Reinforcement Learning

RL is suitable when decisions are sequential, outcomes are uncertain, and actions influence future opportunities. Representative applications are listed below.

1. **Game playing:** tic-tac-toe, chess, Go, Atari, and strategy games.
2. **Robotics:** balance, walking, navigation, grasping, and manipulation.
3. **Autonomous driving:** lane decisions, speed control, and planning under uncertainty.
4. **Industrial control:** refinery control, energy management, and process optimization.
5. **Operations:** scheduling, inventory control, traffic signal control, and resource allocation.
6. **Digital systems:** recommendation, personalization, advertising, and web services.
7. **Healthcare and education:** adaptive interventions, tutoring, and treatment planning, with safety constraints.
8. **Finance:** sequential investment, bidding, and risk-sensitive decision-making.

In each case, the central pattern is the same. The agent must act now, observe what happens, and improve future decisions. A single action cannot be judged only by its immediate effect. It must be judged by the later states and later rewards that it helps to produce.

6. Review of Key Terms

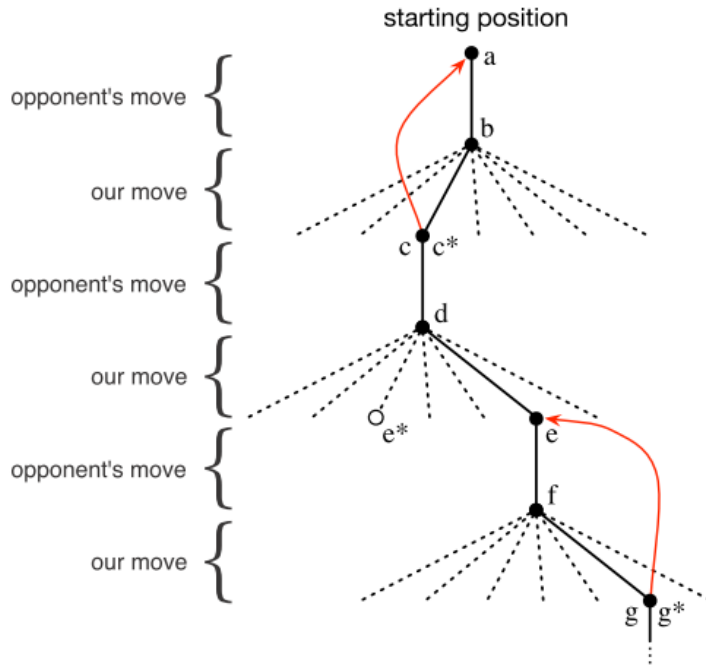
Term	Short definition
Agent	The learner that chooses actions and improves through experience.
Environment	The external system that receives actions and returns next states and rewards.
State, S_t	A representation of the current situation available to the agent at time t .
Action, A_t	A decision or move selected by the agent at time t .
Reward, R_{t+1}	Numerical feedback received after action A_t is taken in state S_t .
Policy, π	The behaviour rule that maps states to actions or action probabilities.
Stochastic policy	A policy that selects actions according to probabilities, written as $\pi(a s)$.
Reward signal	The immediate signal that defines good and bad events for the agent.
Value function	A prediction of expected future reward from a state or state-action pair.
State-value function	A function estimating how good it is to be in a particular state.
Action-value function	A function estimating how good it is to take an action in a state.

Review of Key Terms (continued)

Term	Short definition
Model	A mechanism that predicts how the environment may behave after an action.
Planning	Choosing actions by considering possible future situations before they are experienced.
Model-based method	A method that uses a model of the environment for planning or prediction.
Model-free method	A method that learns directly from experience without an explicit environment model.
Exploration	Trying less-known actions to obtain new information.
Exploitation	Choosing actions currently believed to be best.
Delayed reward	A situation in which consequences of an action appear later, not immediately.
Credit assignment	Deciding which earlier actions contributed to later success or failure.
Temporal-difference error	The difference between successive value estimates used for learning.
Episode	A complete sequence of interaction from a starting state to a terminal state.
Terminal state	A state at which the episode ends, such as a win, loss, or draw in tic-tac-toe.

7. A Toy Example: Tic-Tac-Toe

Tic-tac-toe is a compact example for understanding reinforcement learning because it has **states**, **actions**, **rewards**, **delayed consequences**, and a clear goal. A board position is a state. A legal move is an action. Winning can be treated as reward 1, while losing or drawing can be treated as reward 0 in the simple textbook setup.



X	O	O
O	X	X
		X

A sample board state.

Sequence of considered and selected tic-tac-toe moves.

The value-function approach assigns a number to each possible board state. This number estimates the probability of winning from that state. Initially, winning states may be assigned value 1, losing or drawn terminal states value 0, and other states a neutral value such as 0.5. During play, the agent selects moves that usually lead to states with higher estimated value, while occasionally exploring other moves.

In the diagram, solid lines indicate moves actually taken during a game. Dashed lines indicate moves that were considered but not selected. A move may be selected greedily when it leads to the currently highest-valued next state. Sometimes a non-greedy exploratory move is selected so that the learner can collect useful experience from less-known parts of the game tree.

7.1. Temporal-Difference Learning Rule

Let S_t be the state before a greedy move and S_{t+1} the state after that move. If $V(S_t)$ is the current value estimate of S_t , then the update is

$$V(S_t) \leftarrow V(S_t) + \alpha[V(S_{t+1}) - V(S_t)],$$

where α is the **step-size parameter**. The expression $V(S_{t+1}) - V(S_t)$ is the temporal-difference part of the update. The earlier state's value is moved slightly toward the later state's value.

7.2. Numerical Illustration of the Update

The following cases show how the same rule changes the value of an earlier board state. The values may be read as estimated probabilities of winning from the corresponding board positions.

Case	$V(S_t)$	$V(S_{t+1})$	α	Updated value of $V(S_t)$
Later state looks better	0.50	0.70	0.10	$0.50 + 0.10(0.70 - 0.50) = 0.52$. The earlier state becomes slightly more promising.
Later state looks worse	0.60	0.30	0.10	$0.60 + 0.10(0.30 - 0.60) = 0.57$. The earlier state's value is reduced.
Larger learning step	0.40	0.80	0.25	$0.40 + 0.25(0.80 - 0.40) = 0.50$. A larger α moves the estimate faster.
Close estimates	0.65	0.67	0.10	$0.65 + 0.10(0.67 - 0.65) = 0.652$. Only a small correction is made.

Interpretation. The update does not replace the old estimate fully. It moves it part of the way toward the later estimate. The step-size α controls how large that movement is. Small α gives slow and stable learning. Larger α gives faster but more sensitive learning.

A decreasing step size can make the estimates settle when the opponent is fixed. A step size that does not vanish completely can help the learner continue adapting if the opponent changes slowly. This illustrates why RL often treats learning as an ongoing interaction process rather than a one-time training procedure.

8. Review Questions

1. Define reinforcement learning using the agent-environment interaction framework. Explain the roles of state, action, reward, and cumulative reward.
2. Compare reinforcement learning with supervised learning and unsupervised learning. Clearly distinguish instructive feedback from evaluative feedback.
3. Explain the four main elements of reinforcement learning: policy, reward signal, value function, and model. Why is the value function especially important for long-term decision-making?
4. Discuss the exploration-exploitation trade-off. Why can an agent fail if it only exploits current knowledge or only explores new actions?
5. In the tic-tac-toe example, explain how state values are initialized, how moves are selected, and how the temporal-difference update changes the value of an earlier state.

9. Points to Remember

- RL is a third major learning paradigm, different from supervised and unsupervised learning.
- The agent learns from interaction, not from a fixed dataset of correct answers.
- The reward signal defines the goal, but the value function supports long-term decisions.
- Exploration is necessary because the agent must discover useful actions through experience.
- The temporal-difference rule adjusts an earlier estimate using a later estimate, so learning can happen step by step during interaction.

10. Required Reading

Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press. Required reading: Chapter 1, especially Sections 1.1 to 1.5 covering the definition of reinforcement learning, examples, elements of reinforcement learning, and the tic-tac-toe illustration.

11. Reference

Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). Cambridge, MA: MIT Press.

12. Acknowledgements

These lecture notes are based on and adapted from Chapter 1 of Sutton and Barto's *Reinforcement Learning: An Introduction*, Second Edition.

End of Topic -1